

มหาวิทยาลัยมหิดล
Mahidol University



ITCS215
Introduction to Interactive Multimedia Technology

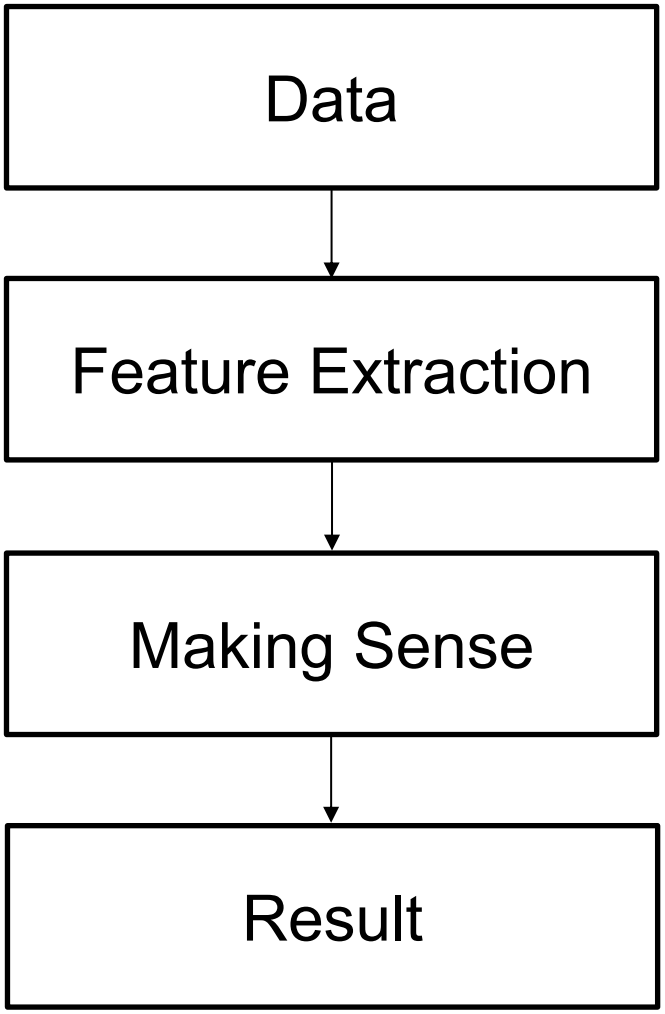
Semester 1/2026

NN/CNN Foundation

Assoc. Prof. Dr. Worapan Kusakunniran

Overview CV Process

Images
Videos
Depths



Color e.g. RGB, HSV, Lab
Histograms
HOG
Edge
Corner
Shape
SIFT
SURF
LBP

Optical flow
STIP

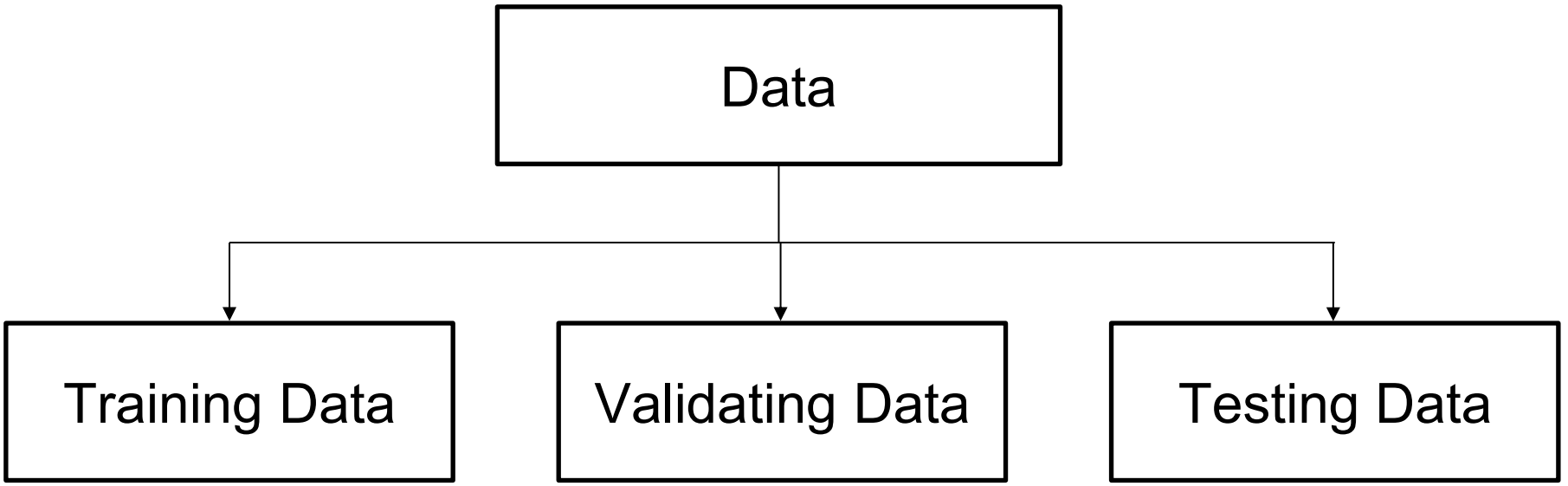
Recognition
Classification
Segmentation
Clustering
Identification
Retrieval
Matching
Tracking

AI
Machine Learning

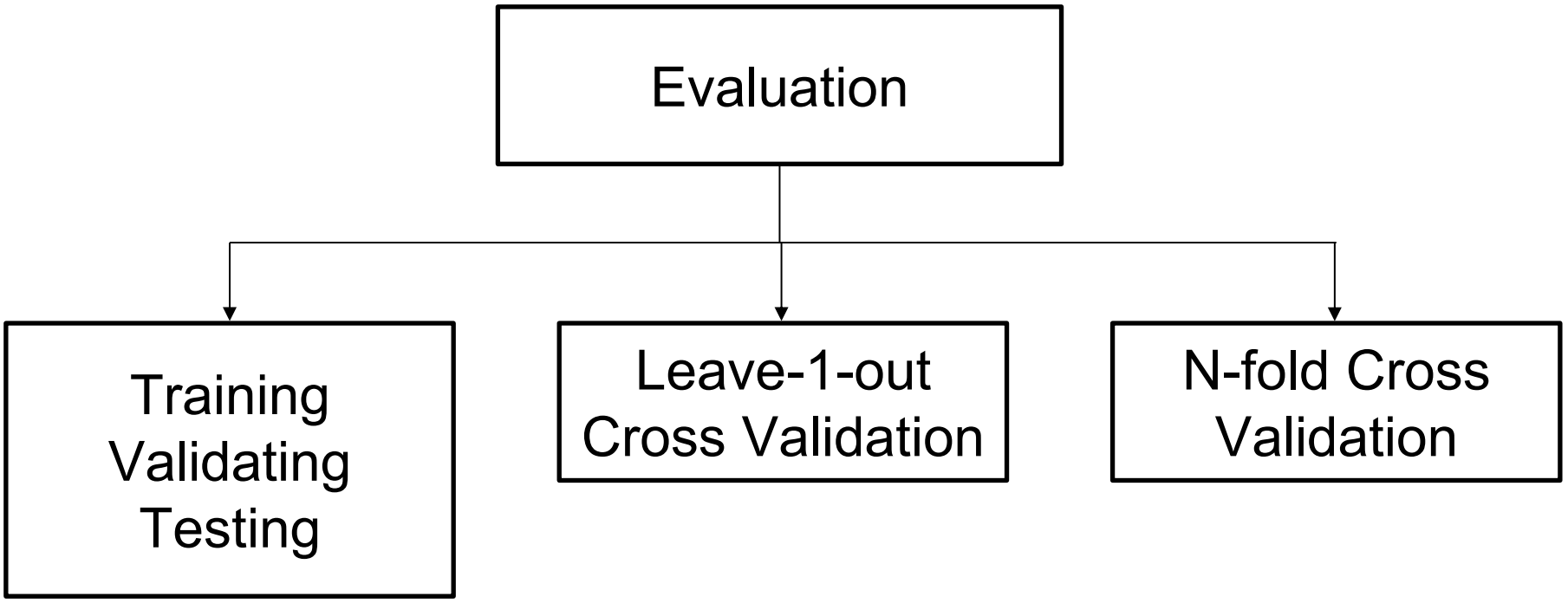
Strong Supervising
Semi Supervising
Non Supervising

ID
Patterns
Model
Class
Interaction

Overview CV Process (cont.)



Overview CV Process (cont.)



Evaluation Measurements

- TP/FP/TN/FP
  - True positive: Sick people correctly identified as sick
  - False positive: Healthy people incorrectly identified as sick
  - True negative: Healthy people correctly identified as healthy
  - False negative: Sick people incorrectly identified as healthy

  - In general, Positive = identified and negative = rejected

  - True positive = correctly identified
  - False positive = incorrectly identified
  - True negative = correctly rejected
  - False negative = incorrectly rejected

		True condition	
	Total population	Condition positive	Condition negative
Predicted condition	Predicted condition positive	**True positive**	**False positive**, Type I error
	Predicted condition negative	**False negative**, Type II error	**True negative**

wikipedia.org

Evaluation Measurements

- Sen/Spec
- Accuracy
- Recall
- Precision

Dice, Jaccard

		True condition	
	Total population	Condition positive	Condition negative
Predicted condition	Predicted condition positive	**True positive**	**False positive**, Type I error
	Predicted condition negative	**False negative**, Type II error	**True negative**

sensitivity, recall, hit rate, or true positive rate (TPR)

$$\mathrm{TPR} = \frac{\mathrm{TP}}{\mathrm{P}} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} = 1 - \mathrm{FNR}$$

specificity, selectivity or true negative rate (TNR)

$$\mathrm{TNR} = \frac{\mathrm{TN}}{\mathrm{N}} = \frac{\mathrm{TN}}{\mathrm{TN} + \mathrm{FP}} = 1 - \mathrm{FPR}$$

precision or positive predictive value (PPV)

$$\mathrm{PPV} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} = 1 - \mathrm{FDR}$$

negative predictive value (NPV)

$$\mathrm{NPV} = \frac{\mathrm{TN}}{\mathrm{TN} + \mathrm{FN}} = 1 - \mathrm{FOR}$$

miss rate or false negative rate (FNR)

$$\mathrm{FNR} = \frac{\mathrm{FN}}{\mathrm{P}} = \frac{\mathrm{FN}}{\mathrm{FN} + \mathrm{TP}} = 1 - \mathrm{TPR}$$

fall-out or false positive rate (FPR)

$$\mathrm{FPR} = \frac{\mathrm{FP}}{\mathrm{N}} = \frac{\mathrm{FP}}{\mathrm{FP} + \mathrm{TN}} = 1 - \mathrm{TNR}$$

false discovery rate (FDR)

$$\mathrm{FDR} = \frac{\mathrm{FP}}{\mathrm{FP} + \mathrm{TP}} = 1 - \mathrm{PPV}$$

false omission rate (FOR)

$$\mathrm{FOR} = \frac{\mathrm{FN}}{\mathrm{FN} + \mathrm{TN}} = 1 - \mathrm{NPV}$$

Threat score (TS) or Critical Success Index (CSI)

$$\mathrm{TS} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN} + \mathrm{FP}}$$

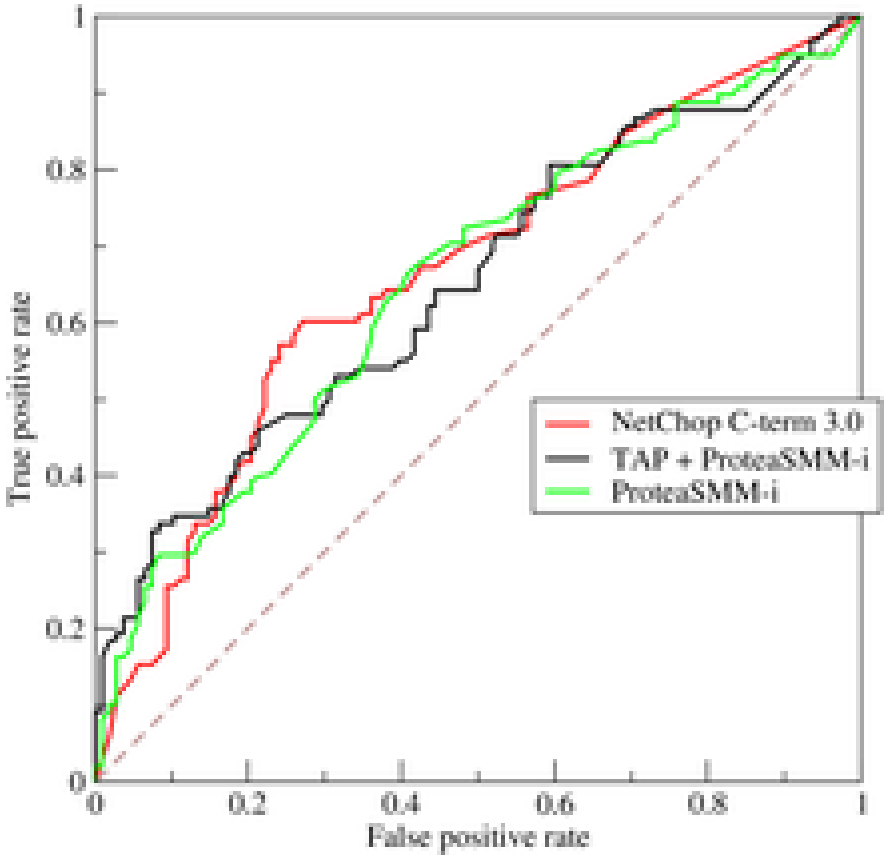
accuracy (ACC)

$$\mathrm{ACC} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{P} + \mathrm{N}} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}}$$

wikipedia.org

Evaluation Measurements

- ROC curve
  - A receiver operating characteristic curve, or ROC curve, is a graphical plot that illustrates the diagnostic ability of a binary classifier system as its discrimination threshold is varied
  - The ROC curve is created by plotting the true positive rate (TPR) against the false positive rate (FPR) at various threshold settings
  - AUC (Area Under Curve)



wikipedia.org

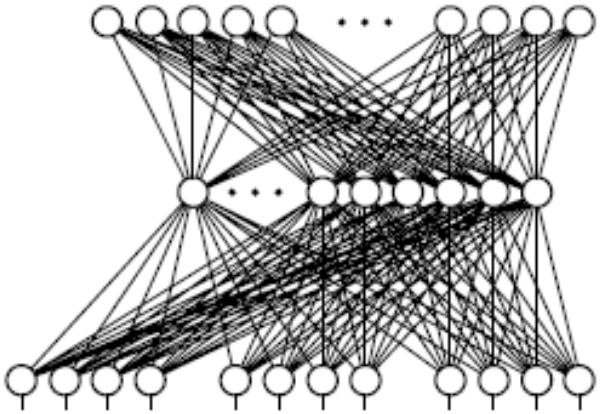
Class Activity 5.1

		True Label	
		DR	No DR
Output class from the developed model	DR	20	50
	No DR	180	1950

- Calculate the performance of the proposed system of DR detection, based on
  - Sen/Spec
  - Accuracy
  - Precision

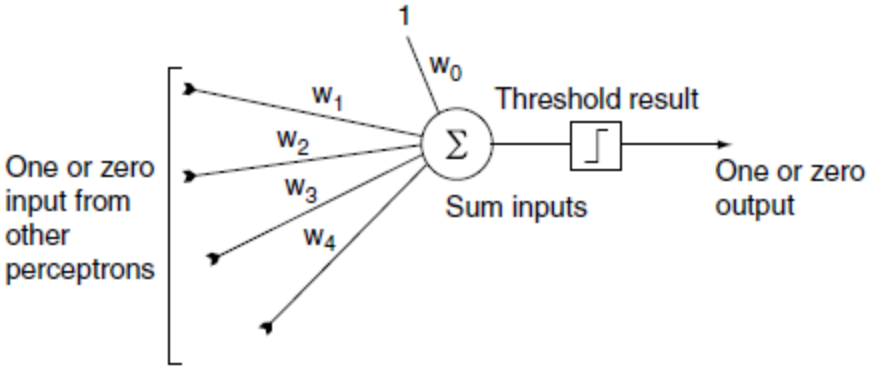
MLP

- Architecture
  - MLP
    - e.g. 3-layers MLP with 1 layer of inputs, 1 hidden layer, 1 layer of outputs
    - n-layers MLP where n > 3 can be achieved by having more hidden layers
    - each neuron connects to all neurons in the layers immediately in font and behind it



MLP (cont.)

- For example: the simple perception algorithm for 0:off/1:on values. E.g. 1 means being a face, 0 means not being a non-face



  - Each input has an associated weight
  - The input values are multiplied by their corresponding weights
  - An additional bias weight is added
  - The final sum is passed through a threshold function
  - If the sum is less than 0 then its output is off, otherwise is on

MLP (cont.)

- Situation:
  - 400 inputs: from 20x20 patch
  - 1 output: being face or non-face
- Architecture
  - 3-layers MLP: input layer, output layer, and hidden layer
  - Input layer has 400 neurons
  - Output layer has 1 neurons
  - Hidden layer should have at least as large as input layer (often much larger)

Class Activity 5.2

- Count a number of learning parameter from this MLP
  - Input: 100 nodes
  - Hidden 1: 200 nodes
  - Output: 10 classes

  - 5.2.1 with no bias weight
  - 5.2.2 with bias weight

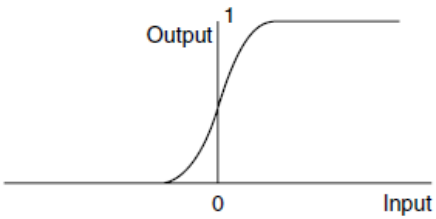
MLP (cont.)

- Framework
  - Initialize all weights in the network with small random values
  - Perform a number of iterations (e.g. hundreds or thousands) of the learning algorithm
  - For each iteration, perform 2 steps
    1. Feed-forwarding
    2. Back-propagation
       - Each example in the training set is selected in turn
  - After finishing all iterations, the learned network is tested using feed-forwarding on the testing set by comparing the returned output with the real output for each testing example.

```
weights
for(each iteration){
    for(each image/batch){
            feed-forwarding
            back-propagation
    }
}
```

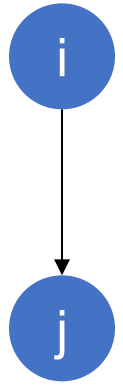
MLP (cont.)

- Feed-forwarding
  - Summing the weighted inputs
    - $x = \sum w_i input_i + w_{bias}$
  - Applying a threshold function to generating its output
    - A popular basis function is a sigmoid function (before, it is a step function)
    - $f(x) = \frac{1}{1+e^{-hx}}$ where h is a parameter controlling the shape of the function
    - The larger h, the nearer to the step function
    - 2 involving factors: the larger number of neurons per layer and the larger size of weights, the smaller h
    - May start with h = 1 and then tuned



MLP (cont.)

- Back-propagation
  - It is a learning process by adjusting weight values
  - w_{ij} is a weight between neurons i and j. It can be updated as



$$w'_{ij} = w_{ij} + \alpha\delta_j o_i$$

where α is gain term or learning rate, δ_j is an error term, and o_i is a value of the neuron i

MLP (cont.)

$$w'_{ij} = w_{ij} + \alpha \delta_j o_i$$

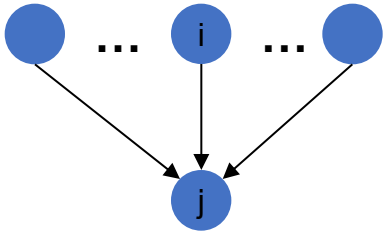
- The gain term
  - or a learning rate
  - Lower gain term
    - More stable learning
  - Higher gain term
    - Faster learning
  - Dynamic gain term ?

MLP (cont.)

$$w'_{ij} = w_{ij} + \alpha \delta_j o_i$$

- The error term
  - If the neuron j is the output node

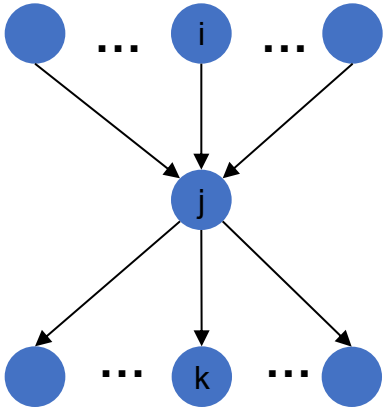
$$\delta_j = o_j(1 - o_j)(t_j - o_j)$$



where o_j is an outputted value of the neuron j, and t_j is a real value of the neuron j

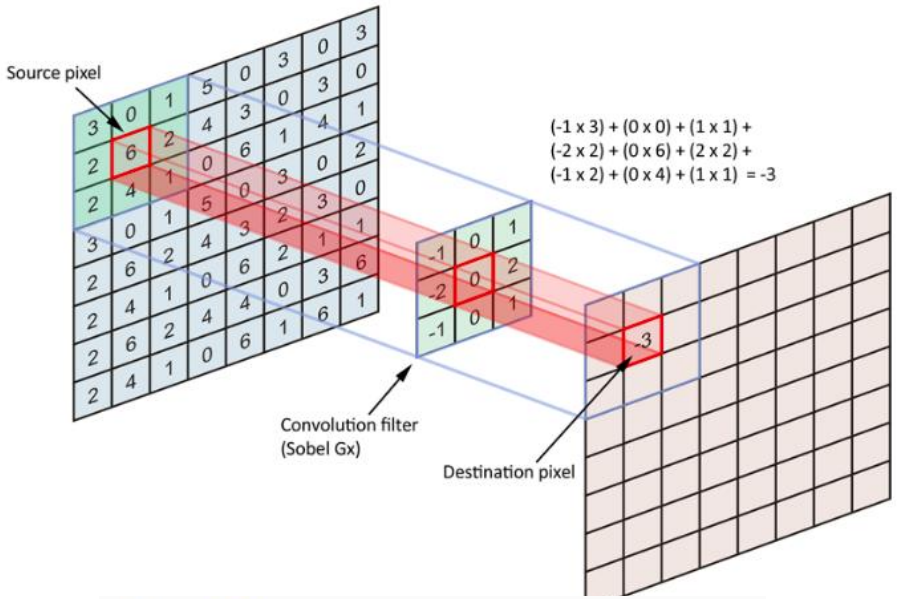
  - If the neuron j is a hidden node

$$\delta_j = o_j(1 - o_j) \sum_k w_{jk}\delta_k$$



CNN

- Key components
  - Convolution
  - Rectified Linear Unit (ReLU)
  - Pooling
  - Fully connected layer



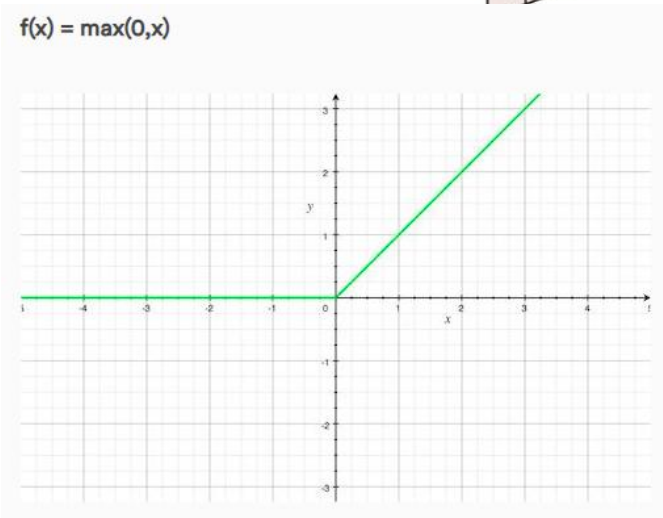
ReLU Layer

Filter 1 Feature Map

9	3	5	-8
-6	2	-3	1
1	3	4	1
3	-4	5	1

→

9	3	5	0
0	2	0	1
1	3	4	1
3	0	5	1



https://hashrocket.com/

https://medium.freecodecamp.org/

CNN

- Rectified Linear Unit (ReLU) – activation function
  - Increase non-linearity in the image
  - Note: Images are naturally non-linear e.g. transition between pixels, borders, colors

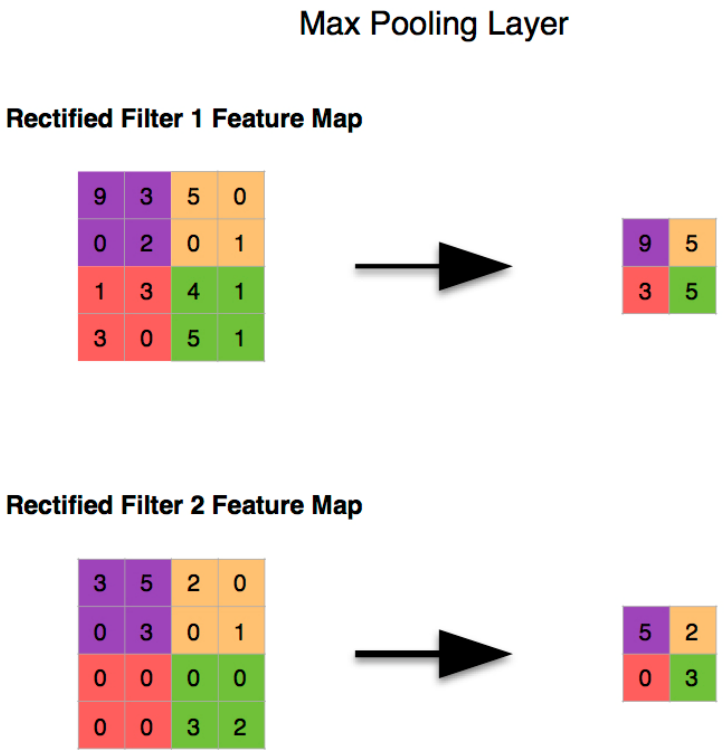




https://www.superdatascience.com/blogs/convolutional-neural-networks-cnn-step-1b-relu-layer/

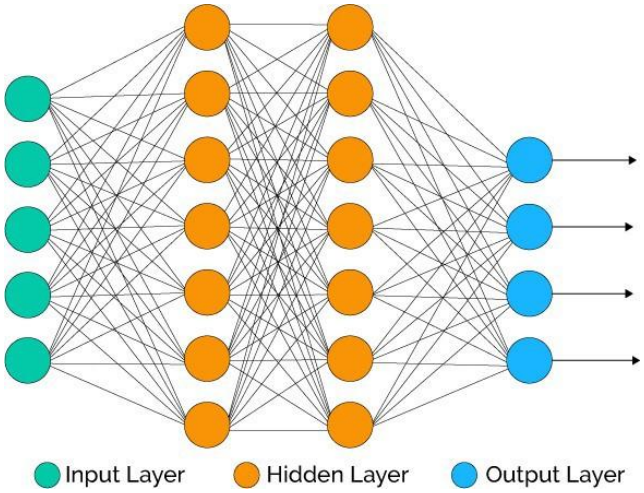
CNN

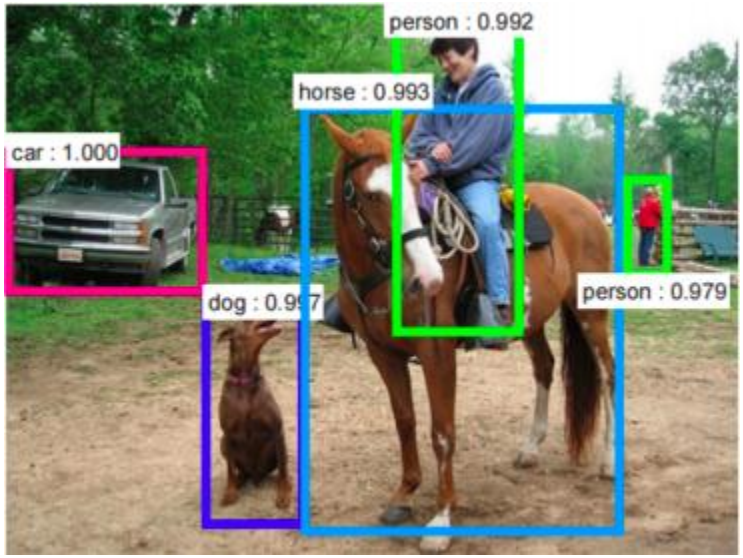
- Key components
  - Convolution
  - Rectified Linear Unit (ReLU)
  - Pooling
    - Max pooling
    - Average pooling
  - Fully connected layer (dense



CNN

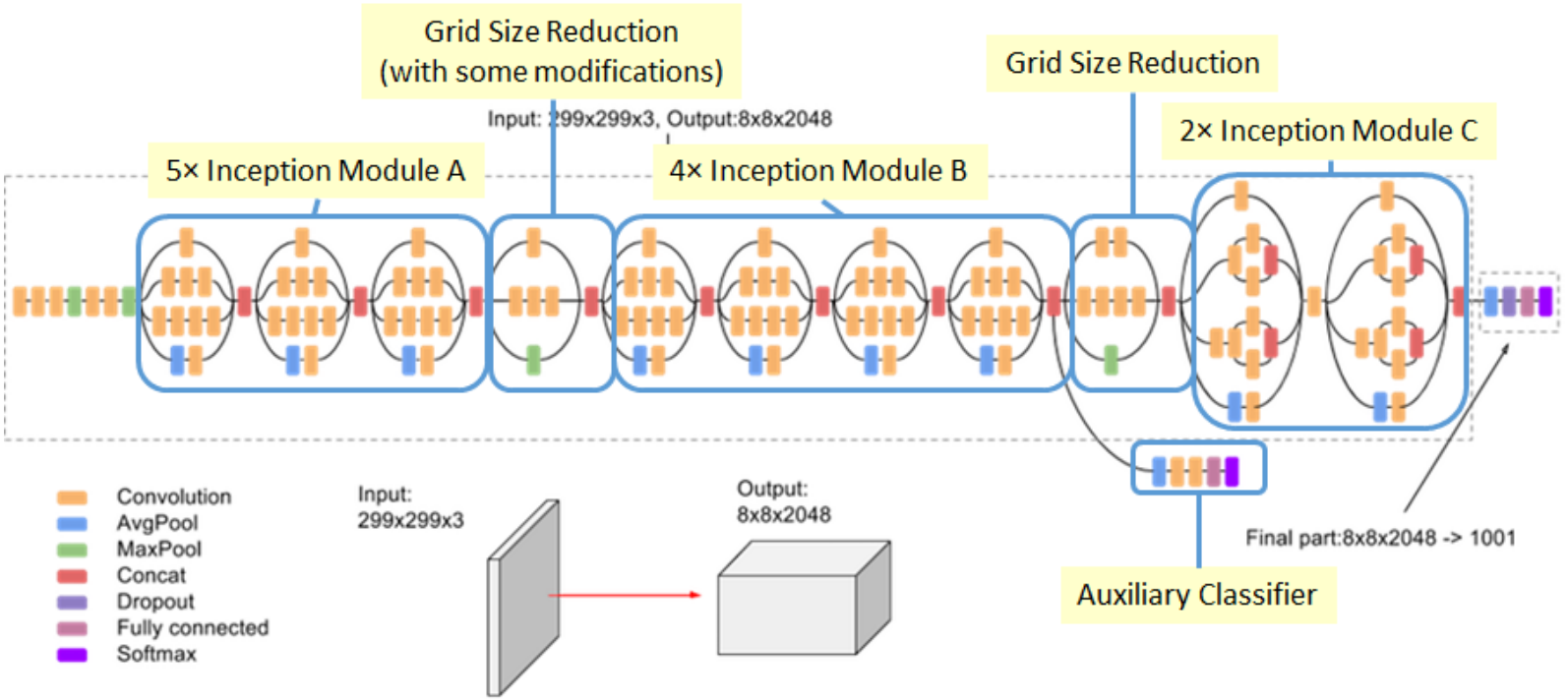
- Key components
  - Convolution
  - Rectified Linear Unit (ReLU)
  - Pooling
  - Fully connected layer





https://towardsdatascience.com/
https://hashrocket.com/
https://medium.freecodecamp.org/

CNN: example



Inception-v3
- **42-layer** deep learning network
- **24M** parameters

https://medium.com/@sh.tsang/review-inception-v3-1st-runner-up-image-classification-in-ilsvrc-2015-17915421f77c

CNN

- Benefits
  - No need of explicit feature extraction
  - Could extract complex patterns
- Challenges
  - Sufficient # of data
  - Sufficient seen variations of training data
  - Overfitting